

NATIONAL OCCUPATIONAL STANDARD FOR METAL CASTERS



NOS.MC.01
FIRST EDITION

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 15th February, 2024.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Manufacturing NOSDT, upon which the following organisations were represented:

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2. Copperbelt University (CBU)
3. Engineering Institution of Zambia (EIZ)
4. Good Time Steel Company Zambia Limited (GTS)
5. Lusaka Vocational and Technical College (LVTC)
6. Makeni Ecumenical Centre (MEC)
7. Ministry of Commerce, Trade and Industry (MCTI)
8. Ministry of Health (MoH)
9. National Institute for Scientific and Industrial Research (NISIR)
10. Northern Technical College (NORTEC)
11. University of Lusaka (UNILUS)
12. University of Zambia (UNZA)
13. Zambia Association of Manufacturers (ZAM)
14. Zambia Forestry College (ZFC)
15. Zambia Qualifications Authority (ZAQA)– Secretariat

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Manufacturing National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Manufacturing sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

A Metal Caster is responsible for manufacturing metal cast parts or other related metal items, as either finished or semi-finished products needed by industry as spare parts, which are critical components in the manufacturing industry. The Metal Caster's role is to manufacture metal castings into parts specified by users using various available alloys. This NOS highlights the core knowledge, skills, competencies, and personal attributes that Metal Casters must possess to be successful in their job.

ACRONYMS AND ABBREVIATIONS

CS	Core Skills
K	Knowledge and Understanding
MC	Metal Casters
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS, and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.MC.01
Occupation	Metal Casting
Job Title	Metal Caster
Job Description	A metal caster is responsible for: • Making patterns and moulds or casts from different materials such as metal, wax, and sand. • Examining moulds to ensure they are clean, smooth and properly coated. • Assembling and embedding cores in the mould using appropriate tools and equipment. • Loading specified amounts of metal flux into furnaces or clay crucibles. • Reading temperature gauges and observing colour changes, adjusting furnace flames, torches or electrical heating units as necessary to melt metal to required specifications. • Operating hand-controlled mechanisms to pour and regulate the flow of the molten metal into moulds to produce castings or ingots. • Turning valves to circulate water through mould cooling system or spraying water on filled moulds to cool and solidify metal. • Transporting metal ingots to storage areas using forklifts.
Job Purpose	A Metal Caster is responsible for using different types of tools, moulds, cores, and furnaces through a casting process to produce parts and tools from molten metal.
ZQF Level	3
Sector	Manufacturing
Sub Sectors	Machine Shops, Metal Fabrication
Other Economic Sector(s) in which the Occupation is Practiced	Mining and Mineral processing; Aviation/Aerospace; Telecommunication; Water and Sanitation.
Other Similar Jobs that can be Performed in the Occupation	Welder, Metal Fabricators.
Minimum Educational Job Entry Qualification(s)	Level 3 Certificate

Practicing License Requirements (if any)	Membership with the Engineering Institution of Zambia at Craftsman or above
Training/RPL	1. Awareness of the Industry Standards, and Rules and Regulations and their application. 2. Quality Enhancement Methods. 3. First Aid Training
Minimum Job Entry Age	20 years
Prior Experience (Suggested)	Minimum of 1 year internship
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Metal Casters are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

The job requires an individual that must be able to read and write a simple report, understand gauges, apply basic mathematics to keep records of inputs and outputs, and plan as well as prioritise their work. The jobholder is expected to be quality-conscious, Occupational Health and Safety orientated, physically fit, courteous, and creative. Further, the individual must have the ability to use fingers, hands, and feet with ease to complete the assigned tasks (dexterity).

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 4 Units, representing the tasks that a jobholder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

Unit 1 [This Unit covers the skills and knowledge required by a Metal Caster to produce casts of various sizes from molten metal]

Unit No.	01
Unit Title	Produce Casts of Various Sizes from Molten Metal
Description	This Unit describes the competencies required to use available tools and mechanisms to produce casts of various sizes from molten metal. The jobholder is responsible for carrying out the casting process from molten metal to a solid cast, while regulating the process parameters, like temperatures, flow rates, and pressures to cool the casting. He/she must be able to use various mechanisms to move the molten alloy and casting from one stage to the other of the casting process.
Scope	This Unit covers the following: • Preparing Patterns and Moulds. • Making Solid Metal from Molten Alloy. • Extract Solid Metal Casting from the Mould. • Finishing.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Preparing Patterns and Moulds	To be competent, the individual must be able to: PC1. Translate information from a drawing or sketch to create a pattern that will determine the mould's shape. PC2. Determine suitable material to make the pattern from. PC3. Use the pattern to make a mould depending on the type of casting. PC4. Use various workshop tools to prepare the mould.
Making Solid Metal from Molten Alloy	To be competent, the individual must be able to; PC5. Receive the molten metal and ensure it is the right alloy for the job, then record it in a log book. PC6. Use the appropriate lifting device or mechanism to move the ladle to the mould. PC7. Operate the hoisting mechanism to pour the molten metal into the mould cavity. PC8. Observe temperature and colour changes to the molten metal to determine the quality of the product. PC9. Monitor and record the temperatures of the molten metal. PC10. Recognise the lay out of the cooling system of the casting process. PC11. Operate water valves, water pumps, and water spray nozzles for cooling and solidifying the cast. PC12. Regulate the water flow using inlet and outlet valves in order to maintain the temperature in the mould for uniform cooling.
Extract Solid Metal Casting from the Mould	To be competent, the individual must be able to: PC13. Identify appropriate tools to extract the cast from the mould. PC14. Extract the solidified metal casting from the mould.

Finishing	To be competent, the individual must be able to: PC15. Inspect the cast for any defects, like shrinkage or cracking, and any other faults. PC16. Measure the dimensions of the casted metal, and ensure the shape comes out as per specifications. PC17. Use appropriate tools to clean the cast by trimming off runners, breaking off the casting gates, grinding, sanding and polishing to give it a final finish as per specification.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company standard operating procedures. OK2. Company code of conduct. OK3. Relevant people and their responsibilities within the work area. OK4. Managing work processes and work plans. OK5. Company safety regulation policy.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Measuring tools, like temperature devices, water flow meters, air pressure gauges, and callipers. TK2. Lifting and hoisting equipment. TK3. Operations of cutting, trimming, and polishing tools. TK4. Types of patterns and moulds. TK5. Basic knowledge of ICT. TK6. Molten alloys, making of solid metal, and chemical composition requirements of casting materials. TK7. Caster inspections, testing, and lock functionality.
C. Regulatory context (knowledge of factory safety rules and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Factories Act and other regulations. RK2. Occupational, Health and Safety Act. RK3. Mine Safety Act and other regulations.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the site. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers. CS3. Record production figures to include input, output, non-conformities, and wastages.
	Reading Skills
	The individual on the job must be able to: CS4. Report production figures to include input, output, and waste.

	CS5. Read and be able to or have the means to give simple instructions in the local language used at the site. CS6. Read and interpret readings of drawings and sketches, temperature, and other gauges or instructions provided for the required job.
	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS7. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS8. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making The individual on the job must be able to: PS1. Determine and report to the superior whether the working space is safe. PS2. Make decisions in emergency situations in the absence of the superior.
	Plan and Organise The individual on the job must be able to: PS3. Plan work and organise required resources in coordination with team members and superiors. PS4. Collect, store, and disseminate updated information.
	Customer Centricity The individual on the job must be able to: PS5. Manage relationships with the supervisor with intent to satisfying their specific requirements.
	Problem-Solving The individual on the job must be able to: PS6. Identify and report the problem to the superiors. PS7. Solve any routine problems that may arise in his/her work.
	Analytical Thinking The individual on the job must be able to: PS8. Identify and report any problem to the superiors.
	Critical Thinking The individual on the job must be able to: PS9. Use common sense and make judgements in day-to-day activities. PS10. Use reasoning skills to Identify and resolve basic problems. PS11. Identify and deal with or report violation of any safety norms which may lead to accidents and report to the superiors.

Unit 2 [This Unit covers the skills and knowledge required by a Master Caster to work according to personal health, safety and environmental requirements at the casting area]

Unit No.	02
Unit Title	Work According to Personal Health, Safety, and Environmental Requirements
Description	This Unit describes the skills and knowledge required to work according to personal health, safety, and environmental rules at the casting area.
Scope	This Unit covers the following: • Follow Organisational Safety Procedures. • Maintain Environmental, Health, and Safety Work Practices. • Implement Good Housekeeping and Environmental Protection Processes and Activities.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Follow Organisational Safety Procedures	To be competent, the individual should be able to: PC1. Identify and report any hazards, risks or breaches in the workplace to the appropriate authorities. PC2. Follow emergency and evacuation procedures in case of accidents, fire incidents or natural calamities. PC3. Adhere to recommended safe practices in handling casting materials, including chemical and other hazardous materials. PC4. Participate in safety awareness programmes, like safety talks and safety demonstrations. PC5. Identify and report near misses, unsafe conditions and acts.
Maintain Environmental, Health, and Safety Work Practices	To be competent, the individual must be able to: PC6. Use appropriate personal protective equipment (PPE) as per work requirements. PC7. Follow safe disposal of waste, harmful and hazardous materials as per the environmental, health, and safety guidelines. PC8. Adhere to safety protocols and practices as laid down by the Environmental, Health and Safety department/team at the site.
Implement Good Housekeeping and Environmental Protection Processes and Activities	To be competent, the individual must be able to: PC9. Collect and deposit waste into identified containers before disposal. PC10. Label and separate containers with toxic or hazardous waste.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company standard operating procedures. OK2. Company code of conduct. OK3. Occupational safety and health policy.

	OK4. People and their responsibilities within the work area.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Procedure for responding to accidents and other emergencies at the site. TK2. Appropriate PPE to be used. TK3. Safe usage of tools, equipment, and materials. TK4. Health and environmental effects of various types of materials. TK5. Storage of waste in appropriate locations, such as: - non-combustible scrap materials and debris; - combustible scrap materials and debris; - general waste and trash (non-toxic and non-hazardous); - sharp edged offcuts, shavings, and mills. TK6. Tools for conducting risk assessment. TK7. Housekeeping activities relevant to a particular task.
C. Regulatory context (knowledge of factory safety rules and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Company safety regulation policy. RK2. Factories Act and other regulations. RK3. Occupational, Health and Safety Act and other regulations.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Fill in safety-related forms for near misses, unsafe conditions, and safety enhancement suggestions. CS2. Note down observations (if any) related to the operations and share the same with the supervisor. CS3. Note down the data for the respective shifts in the log sheets/online systems as per applicability in the organisation. CS4. Prepare requisitions to the supervisor on the requirement of chemicals, equipment, tools, etc.
	Reading Skills
	The individual on the job must be able to: CS5. Read signposts and notice boards relevant to safety. CS6. Read internal information sent by the supervisor/other teams on safety issues.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS7. Speak English (at least working level) and/or have the means to give simple instructions in the local language used at the site on safety issues. CS8. Listen attentively to instructions/communications shared by the site's Environmental, Health and Safety department/team and superiors regarding site safety.

B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Avoid creating unsafe working conditions for others. PS2. Keep the workplace clean and tidy. PS3. Communicate safety problems appropriately to the supervisor and co-workers.
	Plan and Organise
	The individual on the job must be able to: PS4. Plan and organise the safety materials, tools, and equipment required to execute the work.
	Customer Centricity
	The individual on the job must be able to: PS5. Complete all assigned tasks safely, taking into account the safety of the end-users.
	Problem-Solving
	The individual on the job must be able to: PS6. Identify, record, and deal with or report safety risks that may affect one's health and safety, and the environment of others working in the vicinity.
	Analytical Thinking
	The individual on the job must be able to: PS7. Assess and analyse areas which may affect health, safety, and environmental aspects of the workplace.
	Critical Thinking
	The individual on the job must be able to: PS8. Behave and conduct him/herself in a safe manner. PS9. Respond to emergencies as soon as it is safe to do so.

Unit 3 [This Unit covers the skills and knowledge required by a Master Caster to demonstrate in their behaviour how they interact with others in the workplace]

Unit No.	03
Unit Title	Working Effectively with Others
Description	This Unit describes the skills and knowledge required about basic etiquette and competencies that demonstrate behaviour in how interact with others in the workplace.
Scope	This Unit covers the following: Working effectively with co-workers and supervisors.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Working Effectively with Co-workers and Supervisors	To be competent, the individual must be able to: PC1. Display appropriate communication etiquette while Working with others. PC2. Display active listening skills while interacting with Co-workers and supervisors in the workplace. PC3. Demonstrate responsible and disciplined behaviour in the workplace. PC4. Receive information and instructions from the supervisor and co-workers, getting clarification where necessary. PC5. Pass on information to authorised persons who require it and within agreed timelines. PC6. Consult with and assist co-workers to maximise effectiveness and efficiency in carrying out tasks. PC7. Escalate grievances and problems to the appropriate authority as per procedure to resolve and avoid conflict.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of the: OK1. Company standard operating procedures. OK2. Company code of conduct. OK3. Occupational safety and health policy. OK4. Relevant people and their responsibilities within the work area.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Importance of effective communication in the workplace. TK2. Importance of teamwork in organisational and individual success. TK3. Barriers to effective communication. TK4. Company's organisational structure.
C. Regulatory context (knowledge of factory safety)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Company safety regulation policy.

rules and regulations)	RK2. Factories Act and other regulations. RK3. Occupational, Health and Safety Act and other regulations.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills The individual on the job must be able to: CS1. Fill in appropriate technical forms, process charts, and log sheets as per organisational format. CS2. Note down observations (if any) related to the operations and share the same with the supervisor. CS3. Note down the data for the respective shifts in the log sheets/online systems applicable in the organisation. CS4. Prepare requisitions to the supervisor on the requirement of chemicals, tools etc.
	Reading Skills The individual on the job must be able to: CS5. Read and interpret information correctly from various job specification documents, manuals, and health and safety instructions. CS6. Read internal information sent by supervisor or other teams.
	Oral Communication (Listening and Speaking Skills) The individual on the job must be able to: CS7. Check and clarify task-related information. CS8. Liaise with appropriate authorities using the correct protocols. CS9. Communicate with people in a respectful manner, and in line with organisational protocols.
	Decision-Making The individual on the job must be able to: PS1. Identify problems with work planning, procedures, output and behaviour, and their implications. PS2. Prioritise and plan for problem-solving. PS3. Communicate problems appropriately to the supervisor and co- workers. PS4. Identify effective resolution techniques. PS5. Select and apply resolution techniques.
B. Professional Skills	Plan and Organise The individual on the job must be able to: PS6. Plan, prioritise, and schedule work operations as per job requirements. PS7. Organise and analyse information relevant to work.
	Customer Centricity The individual on the job must be able to: PS8. Manage relationships with the supervisor with intent to satisfying their specific requirements.

	Problem-Solving
	The individual on the job must be able to:
	PS9. Identify and report the problem to the superiors.
	Analytical Thinking
	The individual on the job must be able to:
	PS10. Identify and report the problem to the superiors.
	Critical Thinking
	The individual on the job must be able to:
	PS11. Use common sense and make judgements in day-to-day activities. PS12. Use reasoning skills to identify and resolve basic problems. PS13. Identify and deal with or report violations of any safety norms which may lead to accidents and report to the superiors.

Unit 4 [This Unit covers the skills and knowledge required by a Master Caster for Entrepreneurship opportunities in Metal Casting]

Unit No.	04
Unit Title	Entrepreneurship Opportunities in Metal Casting
Description	This Unit describes the skills and knowledge required by a Metal Caster to identify areas for entrepreneurship in Metal Casting.
Scope	This Unit covers the following: Business opportunities in the manufacturing and selling of cast ingots, moulds, cores, and patterns; setting up private foundries and training of new Metal Casters.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Business Opportunities in Manufacturing and Selling; Setting up Private Foundries; Training Metal Casters	To be competent, the individual must be able to: PC1. Identify and explore a wide range of business opportunities available in the metal casting industry. PC2. Develop, organise, and run a new business. PC3. Act on the business idea to disrupt the current market with a new product or service. PC4. Assess the market and demand for the product or service, and the financial risk associated with the business planned.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the standard procedures guiding set up of a business)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant national laws, standards, policies, and procedures followed in the registration of a business or company. OK2. Business/company roles, responsibilities, authorities, and accountabilities.
B. Technical Knowledge	The individual on the job must demonstrate technical knowledge and understanding of: TK1. Creating a business plan. TK2. Acquiring resources and financing for the new business. TK3. Hiring suitable labour for the business. TK4. Providing leadership and management for the business.
C. Regulatory context (knowledge of factory safety rules and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Patents and Company Registration Acts. RK2. Zambia Revenue Authority (ZRA). RK3. Zambia Development Agency (ZDA). RK4. Competition and Consumer Protection Commission (CCPC). RK5. Engineering Institution of Zambia Act.

Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Issue instructions, recommendations, and commendations in writing. CS2. Conduct basic performance assessments and develop performance reports.
	Reading Skills
	The individual on the job must be able to: CS3. Read and also understand basic leadership skills.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS4. Manage basic meetings and discussions. CS5. Provide feedback on strengths, weaknesses, opportunities, and threats (SWOT) for the business. CS6. Give instructions to the team. CS7. Listen attentively and comprehend the information given by the speaker.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Decide whether the business plan will contribute significantly to society by improving the casting methodologies, while assessing the financial risks related to the business.
	Plan and Organise
	The individual on the job must be able to: PS2. Plan, organise, lead, and control business activities. PS3. Use the basic SWOT analysis in the functional areas of responsibility.
	Customer Centricity
	The individual on the job must be able to: PS4. Follow a code of conduct relating to the setting up of the business, and ensure that products or services offered meet the set national standards to meet customer expectations.
	Problem-Solving
	The individual on the job must be able to: PS5. Solve problems within the business. PS6. Identify predictable and unpredictable problems. PS7. Apply appropriate problem-solving techniques and assertiveness in decision-making. PS8. Consult widely to identify possible remedies for complex challenges.

	Analytical Thinking
	The individual on the job must be able to:
	PS9. Analyse and categorise the challenges that affected the setting up of the business.
	PS10. Solve predictable and unpredictable problems quickly and effectively using a step-by-step approach to thinking and breaking down challenges into manageable components.
	Critical Thinking
	The individual on the job must be able to:
	PS11. Use common sense and make sound judgments in day-to-day activities.
	PS12. Use reasoning skills to identify and resolve basic problems.
	PS13. Use intuition to identify any potential problems which could arise during business operations.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to: personal protective equipment, such as leather jackets, safety boots, hardhat, respirators, leather gloves, safety goggles, and shield;

Equipment and Tools:

Casting tools, such as ladles, moulds, and pouring spoons; testing equipment and tools, such as callipers, mass scales, and temperature recorders, hammers, trimming tools, and grinders to polish the end product of a cast.

Raw Materials and Consumables:

Consumables, such as facing sand, graphite, and vegetable oils to prevent sticking of casted parts in the mould; basic molten metal, lubricants, spare parts, and components.

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Metal Casters face challenges, such as exposure to high temperatures, lack of personal protective equipment, inconsistencies in company and government policies and regulations, and pressure from supervisors.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to challenges include: wearing proper safety attire, and ensure participation in safety awareness meetings; adhere to company's policy on safety and operating standards every time; selecting and procuring appropriate equipment and tools for the job; supporting capacity building through training; participating in lobbying and formulation of policies, and allocation of adequate financial resources.

7. WORKING CONDITIONS/ENVIRONMENT

Metal Casters work with a variety of machinery, toxic substances, and volatile materials. Their work environment is susceptible to fires, explosions, fumes, structural failures, and equipment malfunctions. Working conditions include: hot conditions, climbing heights, standing and/or walking for long hours and lifting materials. Metal Casters are also expected to work in day or night shifts, as well as in areas with limited lighting and ventilation.

8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Management, supervisors, subordinates, and other section members.

8.2 External/Outside the Organisation

Government regulators, professional bodies, trainers, consultants and clients, suppliers of consumables, equipment and tools, metal casters from other companies, labour unions, students/interns, occupational health and safety officials.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions, e.g. high temperatures and fumes;
- Be able to walk and stand for long periods of time during casting processes
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy materials e.g. the moulds and cores, tools and equipment;
- Be able to use fingers, hands, and feet with ease to complete the assigned task (dexterity);

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.MC.01		
ZQF Level	3	Version Number	01
Sector	Manufacturing	Date of Approval	February 2024
Sub Sector	Machine Shop and Metal Fabricator	Date of Last Review	N/A
Occupation	Metal Caster	Date of Next Review	February 2029



*“You learn,
We Standardise”*

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